

Audit Report: Loan Payback Prediction ADS

We are auditing an ADS from a Kaggle competition on predicting whether a borrower will fully repay a loan. The goal is to replace manual review processes.

Why we choose this ADS?

- Many models in the competition optimize predictive performance, but overlooked fairness.
- The model uses demographic information such as:
 - Gender, marital status, and education level
- The model is implemented as an ensemble learning model.
- So: Re-evaluate the ADS using fairness metrics and interpretability methods.

[Dataset](#) and [ADS model](#) are accessible through the [Kaggle competition \(loan payback prediction\)](#).

Features

- Original
 - Credit score
 - Interest rate
 - Debt-to-income ratio
- One-hot encoded
 - Gender
 - Marital status
 - Employment status
- Ordinal encode of education level?
- Engineered:
 - loan-to-income ratio,
 - risk score
 - log income

Model Implementation

- 10 candidate classifiers evaluated → LightGBM + CatBoost selected
- 10-fold StratifiedKFold: train on 9 folds, predict on 1
- Per-fold blend:

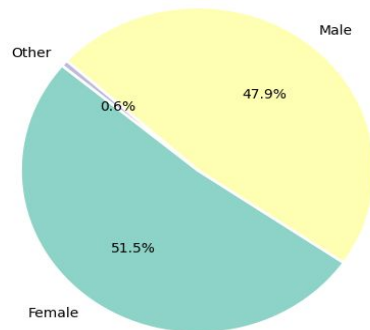
$$\hat{p} = \sqrt{0.70 \cdot p_{LGB}^2 + 0.30 \cdot p_{CAT}^2}$$

- Final prediction: average blended predictions across all 10 folds

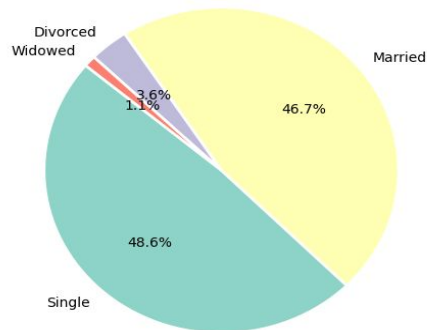
Output: Binary label indicating whether a loan borrower will fully repay the loan.

EDA

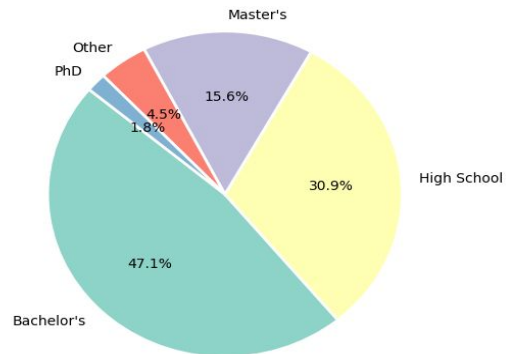
gender Distribution



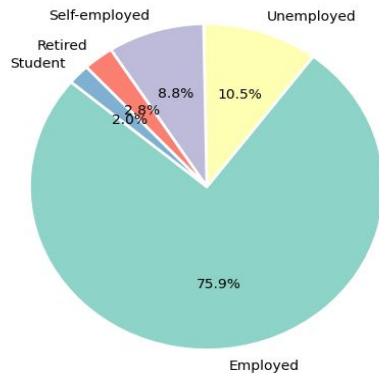
marital_status Distribution



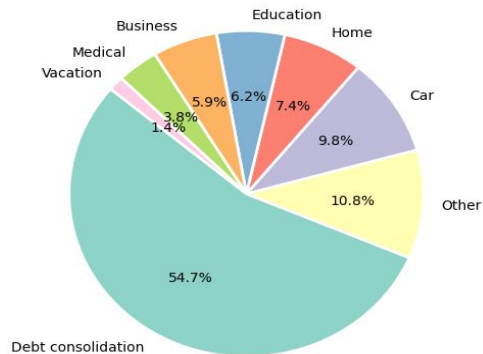
education_level Distribution



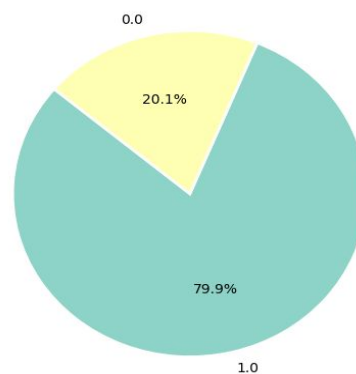
employment_status Distribution



loan_purpose Distribution

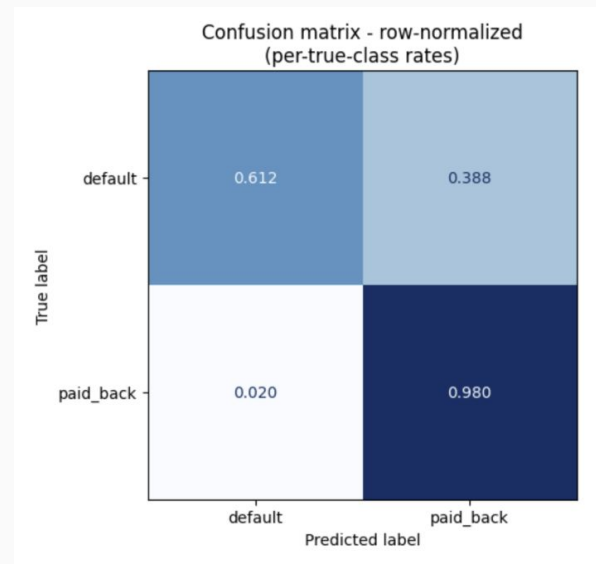


loan_paid_back Distribution



Baseline Performance: A "Loose Approver"

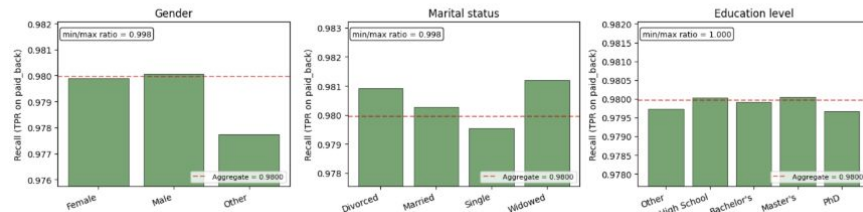
- Accuracy 0.906, ROC-AUC 0.922 (vs. 0.799 base rate)
- But headline numbers hide an asymmetry:
 - Recall on repayers: 0.980
 - Recall on defaulters: 0.612 $\rightarrow$ FPR = 0.388
- Approves 86.1% of all applicants at threshold 0.5
- Takeaway: misses ~39% of true defaulters — the exact failure mode credit models can't have



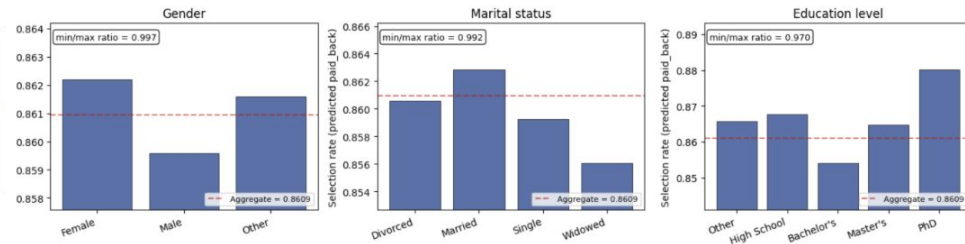
Fairness: Demographic Parity & Equal Opportunity

- Aggregate selection rate: 0.861; per-group rates cluster tightly
- Demographic parity — disparate-impact ratios:
 - Gender 0.997 · Marital 0.992 · Education 0.970
- Equal opportunity — TPR band 0.978–0.981, ratios ≥ 0.998
- Both pass at the single-attribute level. So far so good — but read on.

4.2.2 Equal Opportunity - True positive rate (recall) by group



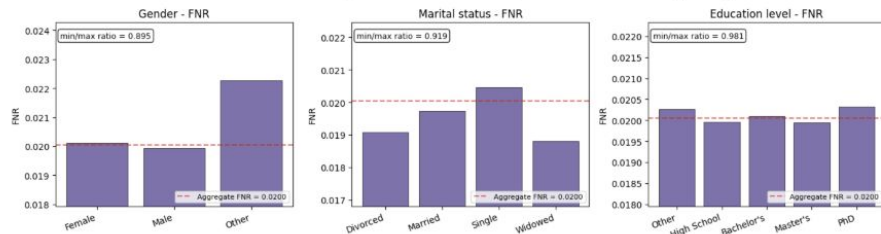
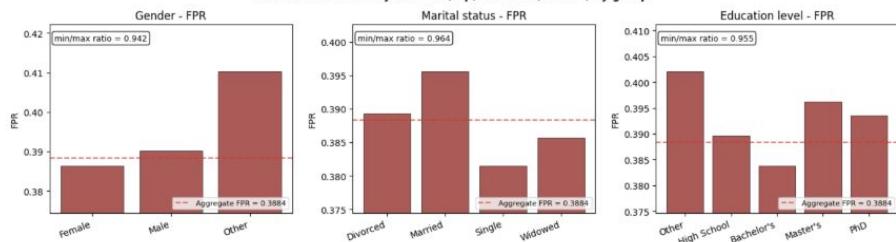
4.2.1 Demographic Parity - Selection rate by group



Error Rates & Intersectional View

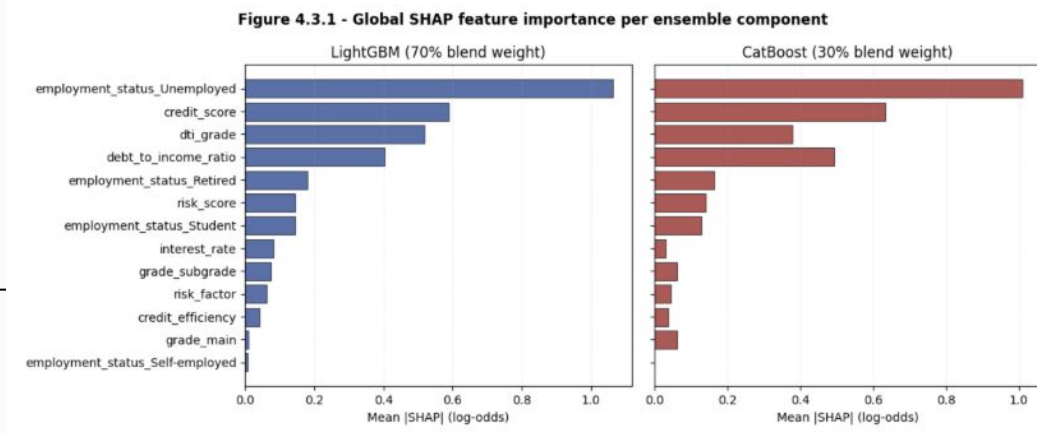
- FPR is where disparities show up:
 - Gender ratio 0.942 (Other = 0.410)
 - Education ratio 0.955 (Other = 0.402)
- FNR essentially uniform (0.019–0.022)
- Intersectional ($n \geq 500$): widest is gender $\times$ education FPR ratio 0.847
- Still clears 4/5ths — but it's the widest gap in the audit

4.2.3 Error Rate Analysis - FPR (top) and FNR (bottom) by group



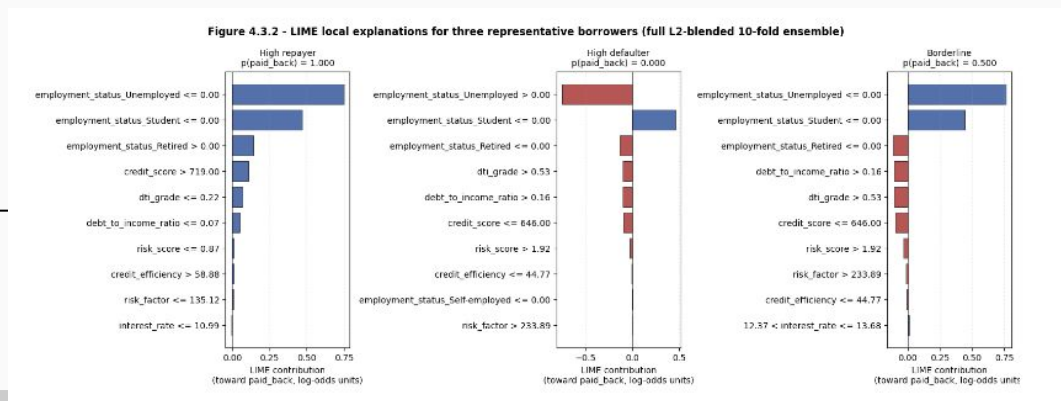
Global Interpretability (SHAP)

- 10-fold SHAP on 2,000-row sample, LightGBM + CatBoost
- employment_status_Unemployed dominates both components
- Grade / subgrade — the engineered risk feature — barely registers
- The model's "credit signal" isn't really a credit signal



Local Interpretability (LIME)

- Three borrowers via ensemble predict_proba:
 - High-confidence repayer ($p=1.00$): unemployed contribution +0.74
 - High-confidence defaulter ($p=0.00$): +0.76
 - Borderline defaulter ($p=0.50$): -0.75
- DTI, credit score $\rightarrow$ an order of magnitude smaller
- SHAP and LIME — different scopes, same conclusion



Findings & Recommendations

- Findings:
- Formal fairness mostly passes (DP, EO, 4/5ths)
- Real concerns:
 - Predictions ride on employment_status proxies, not credit features
 - Loose-approver profile — FPR 0.388 unacceptable for lending
 - Synthetic training data → external validity unverified
- Don't deploy as-is. Before redeployment:
 - Audit & constrain employment-status features
 - Tune prediction threshold cutoff (loose approver)
 - Address class imbalance (weighted training / resampling)
 - Validate on real, non-synthetic data